
Knowing Before Saying: Internal Correctness Signals in Mathematical Reasoning

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Abstract

1 When a language model is confidently wrong, is the error undetected or left
2 unsaid? We look inside the model to find out. Across four open models and
3 four domains (MMLU-Pro, MATH, GPQA-Diamond, and a geometry task
4 graded by a compiler), a simple linear detector reading the model’s internal
5 state predicts whether an answer is correct better than the model’s own
6 stated confidence in 12 of 16 settings, by a mean of +0.09 AUROC. The
7 gap is widest on MATH (+0.20), where an error leaves no visible trace on
8 the page. The detector is not just spotting hard questions: given the same
9 question several times, it still separates the attempts that succeeded from
10 the ones that failed, and read before the model has attempted anything it
11 is at chance. The signal is already there before we ask. At the last token of
12 the answer, with no confidence question in play, correctness is readable on
13 MMLU-Pro at 0.69 against 0.57 from the question alone. And on Mistral
14 the model’s report depends on it: remove that direction after the answer is
15 written and stated confidence can no longer tell the model’s successes from
16 its failures ($0.84 \rightarrow 0.56$), while removing a random direction of the same
17 size changes nothing; turn it up and confidence on wrong answers drops
18 while calibration error halves. Models carry more about their own errors
19 than they say, and on at least one model, what they say depends on it.

20 1 Introduction

21 Ask a language model to solve a competition math problem, then ask how confident it
22 is. It will often say 95, and it will often be wrong. Appendix A follows one such case:
23 Mistral-Small-24B makes a single wrong subtraction inside an otherwise sound derivation,
24 boxes the wrong answer, and then *raises* its stated confidence.

25 This matters for anyone building on these models. A system that is sometimes wrong is
26 workable if it can tell when, because then it can retry, hand off, or abstain. A system that is
27 wrong and sure is not. So the question is whether the confidence number is the whole story,
28 or whether the model knows more than it says.

29 We treat that as two separate questions. Does the model internally represent that an answer
30 is wrong? And does that representation reach the number it reports? We call these *knowing*
31 and *saying*, and the gap between them is what this paper measures.

32 To measure it we read the model’s internal state directly with a simple linear detector, a
33 *probe*, and compare what the probe finds against what the model says. Prior work has shown
34 that a model’s output probabilities are often better calibrated than its words [4], and that
35 whether a given statement is true can be read from activations [1, 2, 5]. We apply those tools

36 to the model’s own attempts, and then go a step further: we intervene on the representation
37 [6] to test whether the model actually uses it.

38 “The model knows” can mean several things, and we test each on its own. That the
39 information is there. That it is about this attempt, not the difficulty of the question. That
40 it exists before anyone asks the model to introspect. That the model’s stated confidence
41 draws on it. And that how much of it reaches speech depends on the model. A probe alone
42 establishes only the first.

43 2 How we look inside

44 **Models and tasks.** Four open models with different architectures: Qwen3.6-27B, GLM-4.7-
45 Flash, Mistral-Small-24B, and Gemma-4-26B. Four domains: MMLU-Pro, MATH, GPQA-
46 Diamond, and GeoGenBench, a geometry task in which the model writes a construction from
47 a natural-language request and a symbolic compiler checks every clause. The compiler is the
48 point. It grades against a formal specification, and a pre-registered human study ($N=200$,
49 two trained raters) found that its disagreements with people ran one way only, lenient and
50 never harsh. When it says `fail`, it is right. Each model answers 150 questions five times per
51 domain, about 750 records per cell, 16 cells in all.

52 **Protocol.** Three turns per problem. We ask the model how confident it is that it will
53 get the problem right, let it attempt the problem, then ask how confident it is that it did.
54 A grader scores the attempt and never tells the model, so any change in stated confidence
55 between the first and third turn is the model’s own judgment, not feedback.

56 **Four readouts of one moment.** At the token where the model is about to write its
57 confidence digit, we record four things: the stated confidence; $P(\text{True})$, the model’s probability
58 of saying “True” when asked whether its answer is correct; the log-probability of the answer;
59 and the model’s internal state, a vector of a few thousand numbers. On that vector we train
60 the probe, a logistic regression that predicts whether the attempt was correct. It is always
61 trained and tested on different questions, and read at one depth chosen in advance, 70% of
62 the way through the network, rather than whichever depth looks best afterward. Our metric
63 is AUROC: the chance that a random correct attempt scores higher than a random wrong
64 one. A value of 0.5 means the readout is useless; 1.0 means it is perfect.

65 **Controls.** The token we read is identical in every record, so at the input layer the probe sits
66 at exactly 0.5; anything it finds deeper was computed during the attempt. A baseline built
67 only from the answer’s length and shape tells us how much the probe adds beyond surface
68 features. A within-question comparison holds difficulty fixed. A deviations log records four
69 bugs these controls caught, which we fixed and reran. One honest note: the original 16-cell
70 matrix was lost to a hardware failure and cannot be recomputed. The four Mistral cells we
71 recaptured reproduce the headline, with a smaller surface margin than the original (+0.21
72 on MMLU-Pro, +0.07 on MATH, −0.11 on geometry, where a failed construction is visibly
73 broken).

74 3 What we found

75 **The model knows more than it says.** The probe beats stated confidence in **12 of 16**
76 cells, one tie, by a mean of +0.09 AUROC (Appendix D). The three losses are small- n cells.
77 The gap is widest on MATH, +0.20, and MATH is where the model’s words go most wrong.
78 On the recaptured Mistral cell, with the grade never revealed, the model raises its confidence
79 after attempting, and raises it *more* on the attempts it got wrong (+22) than on the ones it
80 got right (+16). The probe on those same records reads 0.83. The model’s words move away
81 from the truth while its internal state tracks it.

82 Why MATH? A wrong derivation ends in a clean boxed answer with no visible sign of the
83 error. If stated confidence leans on how polished the output looks, MATH is exactly where
84 that fails. Geometry is the opposite case: a failed construction usually does not compile, the
85 failure is visible on the page, and there the surface baseline does as well as the probe.

Table 1: Mistral, recaptured cells. Left: a probe trained on one domain, tested on another (rows train, columns test; diagonal is in-domain; Geo = GeoGenBench). Right: intervening on the model’s own correctness direction at the confidence token. Gain 1 changes nothing; gain 0 removes the direction; the random row removes a random direction of the same size. Removing the real direction leaves stated confidence unable to separate successes from failures; removing the random one does not.

train ↓ test →	Geo	MMLU-Pro	MATH	GPQA
Geo	0.68	0.67	0.78	0.56
MMLU-Pro	0.70	0.75	0.84	0.59
MATH	0.80	0.72	0.83	0.57
GPQA	0.60	0.69	0.74	0.57

gain	0	1	2	4
confidence, wrong / right	97 / 98	84 / 96	70 / 94	54 / 88
stated AUROC	0.56	0.84	0.85	0.82
random direction, AUROC	0.83	0.84	0.84	unparseable

It is about this attempt, not this question. A probe could succeed just by learning which questions are hard. Two tests rule that out. First, hold the question fixed. Across five attempts at the same question, difficulty cannot vary, yet the probe still separates the successes from the failures (0.72 to 0.74 in representative cells) while the model’s own $P(\text{True})$ falls to near chance (0.47 to 0.59). Second, test the direction itself. Before the attempt, at the last token of the question, the model’s state is identical across all five attempts, so it can encode difficulty but nothing about the outcome. A probe trained after the attempt and read there gives 0.49 on MMLU-Pro, chance, against 0.75 at its own site. The direction that separates good attempts from bad ones does not exist until the model has tried. Geometry is the boundary case: there the pre-attempt state already predicts failure, so that cell is mostly difficulty.

It is there before we ask. Every read so far came from a token the model reaches after being asked to assess itself. Could the question be creating the signal? We read the state at the last token of the answer instead, before any confidence prompt exists. On MMLU-Pro, where the model cannot see failure coming (0.57 from the question alone), correctness is already readable at 0.69 the moment it stops writing. Being asked then sharpens it to 0.77. The signal is made by working; the prompt clarifies it but does not create it. MATH cannot test this, because its question-only read is already 0.81.

One direction, many subjects. A probe trained on one domain and tested on another keeps about 97% of its accuracy on Mistral (0.69 off the diagonal against 0.71 on it, Table 1 left), and 87 to 90% in the original matrix. The model does not have a separate sense of “wrong” per topic. The exception is Gemma-4, whose geometry-trained probes fail on the other domains (0.31 to 0.39) while probes trained elsewhere work on geometry (about 0.70): a probe trained on the one generative task can latch onto features specific to it.

On Mistral, what the model says depends on it. Reading shows the signal is there. To test whether the model uses it, we intervene. At the confidence token, after the answer is written so the answer cannot change, we scale the model’s own projection onto the correctness direction by a gain g , where $g=1$ changes nothing. Remove the direction ($g=0$) and the model’s stated confidence can no longer tell its successes from its failures: 0.84 drops to 0.56, a change of -0.27 $[-0.42, -0.12]$. Remove a random direction of the same size and nothing happens (-0.00 $[-0.03, +0.03]$). This is not damage. The output stays perfectly well formed, and confidence on wrong answers goes *up*, from 84 to 97, until it matches confidence on right answers; the model becomes uniformly sure. Turn the direction up instead, and confidence on wrong answers falls to 70 and then 54 at $g=2$ and 4 while right answers hold near 90; calibration error halves, $0.34 \rightarrow 0.19$ (Table 1 right). Two further tests locate it. The same intervention 30% of the way through the network does nothing; the effect is full by 70%.

122 And a direction learned on MMLU-Pro, where surface features cannot explain the probe,
123 still controls MATH’s report (0.88 $\rightarrow$ 0.71).

124 The same intervention on GLM-4.7-Flash does nothing, even though GLM’s internal signal
125 is the strongest in the study (probe 0.92; within a question, 0.87 against 0.53 for its own
126 words). GLM has the information and its report does not draw on it. That is the clearest
127 case we have of knowing without saying, and it means the causal result is a property of
128 particular models, not of models in general.

129 **What reaches speech depends on the architecture.** A trained probe invites the
130 worry that it has overfit to our data. So we added a second instrument that uses no labels at
131 all: the Jacobian lens [3], fit on generic text, which reads an activation for what it disposes
132 the model to say next. On dense models it matches the probe, 0.82 on MATH and 0.75 on
133 MMLU-Pro on the recaptured Mistral cells against the probe’s 0.83 and 0.75, with a clean
134 input-layer control. Two instruments built from different evidence find the same thing. On
135 the mixture-of-experts and gated-linear-attention models the lens fails (0.31 and 0.43) while
136 the probe still works, and the reason is visible in the lens itself: its “correct” and “wrong”
137 readouts collapse onto each other (cosine 0.96 on MoE against 0.4 on dense). The lens
138 averages over inputs, and these architectures route differently per input. Swap the hybrid
139 for a dense model in the same family and the lens jumps from 0.43 to 0.88. The signal is
140 present in every model; whether a label-free readout can reach it depends on the wiring.

141 4 Discussion

142 Where a task has an exact checker, use it. Where it does not, the probe is the best per-
143 attempt readout we found: keeping only the attempts it scores highest raises accuracy far
144 more than keeping the attempts the model is most confident about, and it still works in cells
145 where the model’s confidence is useless. And because on Mistral the report depends on the
146 direction, the model can be made honest by amplification rather than retraining.

147 **Limitations.** A probe could be reading properties of failed output rather than a dedicated
148 self-assessment; the surface baseline bounds this but does not close it, and the bound is
149 loosest on MATH and geometry (Appendix C). Every number is a single seed. The causal
150 result holds on Mistral and fails on GLM; Gemma-4 and Qwen3.6 are being tested. Whether
151 preference tuning creates the knowing–saying gap is open; a base model would answer it.

152 **Takeaway.** We do not claim that a probe proves introspection. We claim that these models
153 carry information about their own errors that their stated confidence does not report, that
154 the information is about the specific attempt and exists before anyone asks, and that on at
155 least one model the reported number depends on it. A confidence score is not the whole
156 story, and the rest of the story is readable.

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A The elicitation protocol, and one record end to end

This appendix gives the exact prompts behind every record, then one complete record from the Mistral-Small-24B $\times$ MATH cell, summarised first and then given in full.

One record, Mistral-Small-24B $\times$ MATH. *What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$? The model sets the problem up correctly and gets $18 \times 622953 = 11213154$ right, then writes $r = -11213141 + 11213154 = 3$ and answers 3. That subtraction gives 13, which is the gold answer. Stated confidence: 90 before the attempt, 95 after it. Other readouts of the same attempt: $P(\text{True})$ 0.32, mean answer log-prob -0.05 .*

A single arithmetic slip inside a fluent derivation. Having just made the error, the model revises its stated confidence upward, while $P(\text{True})$ on the same transcript assigns only 0.32 probability to correctness. The rest of this appendix gives the prompts and the full transcript.

A.1 Prompts

Each record is four forward passes. A read is first taken at the last prompt token of the bare question, before any elicitation, then three turns are generated. The confidence turns run with thinking off so the read lands on a clean one-line reply; the attempt turn runs with thinking on for hybrid reasoners. The fixed **Confidence:** N format is what makes the read site content-neutral: the digit token has identical local context on every record, so layer-0 activations carry no signal by construction.

System prompt (MATH).

Solve the problem step by step. Put your final answer in `\boxed{}`.

Turn 1, prospective confidence. The problem is followed by:

Before answering, estimate how likely you are to answer this question correctly. Reply with EXACTLY one line and nothing else:
Confidence: N
where N is an integer from 0 (certain to be wrong) to 100 (certain to be correct). Do NOT answer yet.

Turn 2, the attempt. The turn-1 reply is left in context as the assistant message, then:

Now give your final answer inside `\boxed{}`.

Turn 3, retrospective confidence. The attempt is left in context, then:

Now assess the answer you just gave. On a scale of 0 to 100, how confident are you that it is correct? Reply with exactly one line and nothing else:
Confidence: N

P(True) baseline. A separate teacher-forced branch off turn 2, no generation. Next-token mass on True versus False:

Consider your final answer above. Is it correct? Reply with exactly one word:
True or False.

212 Grading is external and is never shown to the model, so the turn-3 revision is blind self-
 213 assessment.

214 A.2 One record, verbatim

215 Record `math_test_number_theory_753`, sample 2 of 5.

216 **Problem.** *What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$?*

217 **Turn 1 reply.** Confidence: 90

218 **Turn 2 reply (complete).**

219 First, we need to find the remainder when -11213141 is divided by 18.

220 1. Calculate $-11213141 \bmod 18$:

$$-11213141 \div 18 \approx -622952.2778$$

221 The integer part of the division is -622953 .

$$-11213141 = 18 \times (-622953) + r$$

222 where r is the remainder.

223 2. Calculate the remainder r :

$$r = -11213141 - 18 \times (-622953)$$

$$r = -11213141 + 11213154$$

$$r = 3$$

226 Therefore,

$$-11213141 \equiv 3 \pmod{18}$$

227 So, the integer n that satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$ is 3.

228 **Turn 3 reply.** Confidence: 95

229 **What happened.** The method is right and every intermediate is right: $18 \times 622953 =$
 230 11213154 checks out, and the model has written the correct subtraction on the line above its
 231 answer. It then evaluates $-11213141 + 11213154$ as 3 rather than 13. Gold is 13. Nothing
 232 in the presentation marks the slip. The derivation is well-formed, correctly structured, and
 233 confidently laid out, which is precisely why stated confidence rises from 90 to 95 across an
 234 attempt that failed.

235 **Readouts of this attempt.**

readout	value
stated confidence, prospective (turn 1)	90
stated confidence, retrospective (turn 3)	95
P(True)	0.32
mean answer log-probability	-0.05
graded outcome	incorrect

237 This is one record, chosen to illustrate the failure mode rather than to carry evidential weight;
 238 the cell-level numbers are in Table 2.

239 B Calibration beyond ECE, and cell-level multiplicity

240 ECE depends on binning and on sample composition, so we report two binning-free proper
 241 scoring rules alongside it. On Mistral $\times$ MATH (750 records, accuracy 0.648), the model’s
 242 retrospective stated confidence gives Brier 0.262 and NLL 1.470 against a 10-bin ECE of
 243 0.271. P(True) on the same records gives Brier 0.386: better ordered than stated confidence,

worse calibrated in absolute terms. The steering results in Table 1 are reported as ECE because the steered runs stored per-gain aggregates rather than per-record confidences; recomputing Brier and NLL at each gain requires re-running the intervention and is the first thing we would add.

The cell-level bootstrap counts in Table 2 are uncorrected. With 16 cells, some individually significant results are expected under the null, so the cell counts should be read as descriptive. The aggregate pattern is the claim that carries weight: the effect is positive in 12 of 16 cells with one tie, significantly positive in 10, and never significantly negative. A hierarchical model pooling across cells with model and domain as random effects is the right test and is not yet run.

C Alternative explanations for the probe signal

A probe could key on properties of failed derivations rather than on a dedicated metacognitive variable: low token probability, hedging language, or, on MATH, an early-layer channel next to the answer text. Our surface-features baseline (answer length, digit count, line count, word count, brace count) bounds this without closing it, and the bound is domain-dependent: the probe adds +0.21 over surface on MMLU-Pro but only +0.07 on MATH and −0.11 on geometry, where a construction that fails to compile is visibly malformed. Two results argue against a pure surface account on the domains where the signal is strong. A direction fitted on MMLU-Pro, where surface reaches only 0.54 against a probe of 0.75, still governs MATH’s self-report under ablation (0.880 → 0.709). And the pre-attempt read, taken where the activation is byte-identical across a question’s attempts, cannot recover the post-attempt direction on MMLU-Pro (0.490, chance), which a surface-of-the-question account would predict it could. A linear probe at one token also undercounts nonlinear or multi-token signal, so the reported numbers are a floor.

D Per-cell numbers and specificity controls

Table 2: Post-attempt correctness discrimination (AUROC). Left: stated confidence vs. the probe in representative cells; overall the probe wins 12 of 16 with one tie, mean +0.09, with a paired bootstrap significantly positive in 10 of 16 and never significantly negative, uncorrected at the cell level (Appendix B). Right: Mistral specificity controls. “pre” is the last prompt token before the attempt; “post” is after the attempt; “resid.” removes the pre score; “post→pre” applies that fitted post readout at the pre-attempt site, where only difficulty is visible.

model × domain	stated	probe	Mistral	pre	post	resid.	post→pre
Gemma-4 × MATH	0.57	0.78	MMLU-Pro	0.568	0.754	0.751	0.490
Qwen3.6 × MMLU-Pro	0.67	0.84	MATH	0.792	0.834	0.701	0.624
GLM × MMLU-Pro	0.67	0.85	GPQA	0.593	0.572	0.544	0.479
Mistral × MATH	0.79	0.83	GeoGen	0.770	0.676	0.505	0.648